Thm On a Riemannian mfld M, there is a unique connection ∇ on TM such that

1) $\nabla_X Y - \nabla_Y X = [X, Y]$ (symmetry)

2) $X g(Y,Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$ (compatibility)

(This is called the "Riemannian connection")

Proof: Use (1) and (2) to express

$$X g(Y,Z) + Y g(X,Z) - Z g(X,Y) = 2 g(\nabla_X Y, Z) + \text{terms involving only } [\cdot,\cdot]$$

$\Longrightarrow$ we can solve for $g(\nabla_X Y, Z)$ in terms of g and $[\cdot,\cdot]$

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Riemannian connection in local coord's:

define: Christoffel symbols Γ_{ij}^k are the coeffs so that

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k$$

(Γ_{ij}^k depends on coord's, not a tensor)

compute: $\partial_i g_{jh} = \partial_i g(\partial_j, \partial_h)$

$= g(\nabla_{\partial_i} \partial_j, \partial_h) + g(\partial_j, \nabla_{\partial_i} \partial_h)$ by compatibility

$= \Gamma_{ij}^l g_{lh} + \Gamma_{ih}^l g_{lh}$

Solve for Γ_{ij}^k:

$$\boxed{\Gamma_{ij}^k = \frac{1}{2} g^{kl} \left(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij} \right)}$$

Example: cone $\{z = ar\}$ in $\mathbb{R}^3$

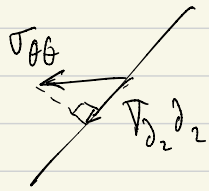
$$\sigma(z,\theta) = (az\cos\theta, az\sin\theta, z)$$

Compute $\nabla_{\partial_2}(\partial_2)$:

1) geometrically $\quad \partial_2 = \sigma_\theta = (-az\sin\theta, az\cos\theta, 0)$

$$\frac{\partial}{\partial\theta}\sigma_\theta = (-az\cos\theta, -az\sin\theta, 0) \qquad \text{(not tangent)}$$

points towards z-axis



$= $ projection $/\!/$ to σ_z

$$e_1 := \frac{\sigma_z}{\|\sigma_z\|} = \frac{(a\cos\theta, a\sin\theta, 1)}{\sqrt{1+a^2}}$$

$$\nabla_{\partial_2}\partial_2 = (\sigma_{\theta\theta}\cdot e_1)\, e_1$$

$$= \frac{-a^2 z}{\sqrt{1+a^2}}\, e_1$$

$$= \frac{-a^2 z}{1+a^2}\,\partial_1$$

2) Christoffel $\quad [g_{ij}] = \begin{pmatrix} 1+a^2 & 0 \\ 0 & a^2 z^2 \end{pmatrix}$

$$\nabla_{\partial_2}\partial_2 = \Gamma_{22}^i \partial_i$$

$$\Gamma_{22}^1 = \frac{1}{2} g^{ij}\left(2\partial_2 g_{2j} - \partial_j g_{22}\right)$$

$$\Rightarrow \Gamma_{22}^1 = \frac{1}{2}\frac{1}{1+a^2}\left(-\partial_z(a^2z^2)\right) = \frac{-a^2 z}{1+a^2}$$

$$\Gamma_{22}^2 = 0$$

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Thm If $M \subset \mathbb{R}^N$ and g is the restriction of the Euclidean metric, then the Riem. connection agrees with the projection into tangent spaces

Covariant differentiation

Let $\gamma : I \to M$ be a smooth curve and X a vector field along γ

i.e. $X(t) \in T_{\gamma(t)} M$ and varies smoothly

defines the covariant derivative of X along γ is

$$D_t X := \nabla_{\dot\gamma(t)} X(t) \in T_{\gamma(t)} M$$

In coords, $\gamma = (\gamma^1, \dots, \gamma^n)$ and $X(t) = a^i(t)\,\partial_i$

$$\dot\gamma = \dot\gamma^j \partial_j \quad \text{or} \quad \dot\gamma f(t) = \dot f(t)$$

$$\begin{aligned} D_t X &= \nabla_{\dot\gamma} (a^k \partial_k) \\ &= \dot a^k \partial_k + a^k \nabla_{\dot\gamma} \partial_k \\ &= \dot a^k \partial_k + a^k \dot\gamma^j \nabla_{\partial_j} \partial_k \\ &= \dot a^k \partial_k + a^k \dot\gamma^j \Gamma^i_{jk} \partial_i \end{aligned}$$

$$D_t X = \left[\dot a^i + \dot\gamma^j a^k \Gamma^i_{jk} \right] \partial_i$$

def: γ is <u>geodesic</u> if $D_t \dot\gamma = 0$ (covariant acceleration)

in coords $\ddot\gamma^i + \dot\gamma^j \dot\gamma^k \Gamma^i_{jk} = 0$ for $i = 1, \dots, n$

(linear system of 2nd order ODE)

<u>Thm</u> A unique geodesic exists given initial conditions $\gamma(0)$ and $\dot\gamma(0)$.

Parallel transport

A vector field X along a curve γ is <u>parallel</u> if

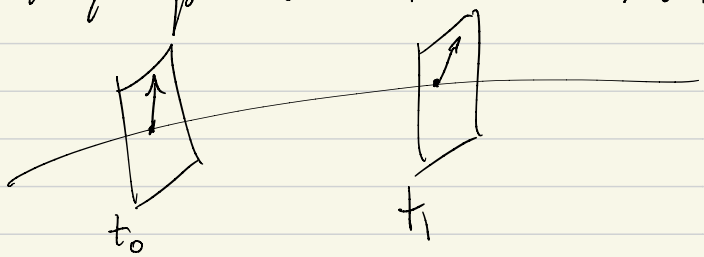
$$D_t X = 0$$

In coord's, $X = a^i \partial_i$ is parallel if

$$\dot{a}^i + a^j \dot{\gamma}^k \Gamma^i_{jk} = 0 \quad \text{for } i = 1, \dots, n$$

(linear 1st order system of ODE)

<u>Thm</u> Given a curve γ and starting vector $X(0) \in T_{\gamma(0)}M$,
$\exists$ unique parallel vector field $X(t)$.



Parallel transport defines a map $T_{\gamma(0)}M \longrightarrow T_{\gamma(1)}M$,
"connects" the tangent planes along γ